

EEE 306 (July 2023)

Power System I Laboratory

Final Project Report

Section: B1 Group: 05

Fast-Decoupled Load Flow (FDLF) Study for a System with Any Number of Bus

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1 Abstract

This project focuses on the implementation and study of the Fast-Decoupled Load Flow (FDLF) algorithm for power flow analysis in electrical systems independent of the number of buses. The FDLF algorithm is implemented using MATLAB to efficiently analyze systems with any number of buses. IEEE standard test cases are utilized, and comparisons are done with Power System Analysis Framework (PSAF) to validate the accuracy of the implemented method. Results illustrate the effectiveness of the FDLF algorithm in predicting voltages and angles, with slight discrepancies detected in reactive power calculations as the number of buses grows. This method reduces the dependency of load flow study on reactive power calculations. Through a combination of theoretical analysis, algorithm development, and simulation experiments, the project intends to elucidate the effectiveness and practicality of FDLFS in real-world scenarios.

2 Introduction

Performing load-flow calculations is crucial in various aspects of power system analysis, including system planning, operational planning, and operation/control. Selecting an appropriate solution method for practical applications poses a challenge, requiring a careful evaluation of methods based on factors such as storage, speed, and convergence characteristics. Each method has its strengths and weaknesses, and the choice depends on specific application requirements and available computing resources.

Newton-Raphson method is widely used for routine solutions due to its popularity and effectiveness. However, it may be limited for small-core applications, where the Gauss-Seidel-type method is more economical. Newer methods are faster for approximate repetitive solutions, such as those needed in AC security monitoring. Numerical methods are most efficient when they exploit the physical properties of the system being solved.

Efficient solutions often leverage network sparsity through ordered elimination and skilled programming in methods like Newton-Raphson. Recent developments involve exploiting the loose physical interaction between MW and MVAR flows in a power system by mathematically decoupling MW-F and MVAR-V calculations.

The method discussed in this report integrates these ideas, combining the advantages of existing methods. It is simpler, faster, and more reliable than Newton-Raphson method, with lower storage requirements for entirely in-core solutions. Its core requirements are similar to the Gauss-Seidel method, making it suitable for accurate load flows and outage-contingency evaluation studies, whether performed on- or off-line.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of this project comprises the implementation and study of the Fast-Decoupled Load Flow (FDLF) algorithm for power flow analysis in practical electrical systems. Specifically, the project seeks to produce a MATLAB code capable of efficiently analyzing and optimizing systems with any number of buses using the FDLF algorithm. The investigation will entail showing voltages and angles in the system and comparing the results with existing power system modeling software (PSAF). The primary goal is to examine the accuracy and computational efficiency of the FDLF algorithm in practical power systems.

3.1.2 Literature Review

Load-flow calculations are fundamental in power system analysis, aiding in system planning, operational decision-making, and control. Researchers have explored various solution methods, each with its merits and limitations. Here's a summary of relevant literature:

1. ["Fast Decoupled Load Flow Method"- B. Scott, O. Alsac ¹:](#)
 - Provided calculations for networks of any size, and can be implemented efficiently on computers with restrictive core-store capacities.
 - Development on other work employing the MW- Θ / MVAR-V decoupling principle, and its precise algorithmic form had been determined by extensive numerical studies.
 - The paper gave details of the method's performance on a series of practical problems of up to 1080 buses
 - A solution to within 0.01 MW/MVAR maximum bus mismatches was obtained in 4 to 7 iterations, each iteration being equal in speed to 1½ Gauss-Seidel iterations or 1/5th of a Newton iteration.
 - Correlations of general interest between the power-mismatch convergence criterion and actual solution accuracy were obtained.
2. ["Fast Decoupled Load Flow: Hypothesis, Derivations, and Testing" - A. Monticelli, A. Garcia, O. R. Saavedra ²:](#)
 - Two algorithms are introduced – primal and dual. The primal algorithm leads to BX version of the FDLF where matrix B' is the Jacobian submatrix H computed at $V = 1 \text{ p.u.}$, $\theta = 0^\circ$. The dual algorithm shares some of the features of the standard FDLF, but was different in some critical aspects where matrix B' was given by the Jacobian submatrix L computed at flat start.
 - Version XB (standard FDLF) required extra approximations as that involved ignoring the effect of the PV buses and shunts on the $P - \theta$ sensitivities.
 - Results of the tests had been carried out using 10^{-4} p.u. convergence tolerances strict successive iteration scheme, and flat start initialization.

- The primal method had presented better performance than Standard method in nearly all the cases for the tests performed for a variety of operating conditions.

3. “Designs of Fast Decoupled Load Flow for Study Purpose” – Yubin Yao, Mingwu Li ³:

- Employed sparse matrix techniques to greatly reduce computer storage and time requirements.
- To make the program easy to use and modify, coefficient matrices B' and B'' of the admittance matrix B in that paper were both stored with an $n \times n$ arrays.
- A real network named Northeast network in China was used to testify conclusions. There were 445 buses and 544 branches (including transmission lines and transformers) in that. The specified convergence tolerance was 0.0001.

In summary, FDLF combines simplicity, speed, and reliability, making it a valuable tool for load-flow analysis. Researchers continue to refine and adapt FDLF to meet evolving power system needs.

1. <https://ieeexplore.ieee.org/document/4075431>
2. https://www.academia.edu/27006861/Fast_decoupled_load_flow_hypothesis_derivations_and_testing
3. https://www.researchgate.net/publication/257711931_Designs_of_Fast_Decoupled_Load_Flow_for_Study_Purpose/fulltext/026a47830cf2db63b3ab1b85/Designs-of-Fast-Decoupled-Load-Flow-for-Study-Purpose.pdf

3.1.3 Formulation of Problem

The problem formulation requires using the FDLF study in MATLAB to solve the power flow equations for systems with any number of buses. The mathematical approach of the problem entails modelling the system's network using nodal admittance matrix (Y_{bus}) and solving the power flow equations iteratively to obtain the voltages and angles at each bus. Additionally, the problem formulation comprises developing IEEE standard test cases for validation and comparative purposes.

The FDLF algorithm involves solving a set of nonlinear power flow equations iteratively to obtain the system's voltages and angles. This process is mathematically represented by the following equations:

$S = V \cdot \text{conj}(Y_{bus} \cdot V)$ - Equation representing the power injections at each bus.

$S1 = S - S_{sp}$ - Equation representing the mismatch between scheduled and actual power injections.

$dP = \text{real}(S1(NS))$ - Equation representing the real power mismatch at PV and PQ buses.

$dQ = \text{imag}(S1(PQ))$ - Equation representing the reactive power mismatch at PQ buses.

$B1 = -\text{imag}(Y_{bus}(NS, NS))$ - Equation representing the imaginary part of the diagonal

elements of the nodal admittance matrix for PV and PQ buses.

$B_{11} = -\text{imag}(Y_{\text{bus}}(PQ, PQ))$ - Equation representing the imaginary part of the diagonal elements of the nodal admittance matrix for PQ buses.

$d\theta = -\text{inv}(B_{11}) \cdot dP$ - Equation representing the correction vector for voltage angles.

$dV = -\text{inv}(B_{11}) \cdot dQ$ - Equation representing the correction vector for voltage magnitudes.

These equations form the basis of the iterative solution process employed in the FDLF algorithm to converge to a solution that satisfies the power flow equations.

Now, in this project, we intend to find how beneficial the FDLF would be when it comes to handle power systems varying in size and complexity. Additionally, knowing the efficiency of calculations in environments with restrictive core-store capacities, enabling efficient implementation on computers with limited resources.

3.1.4 Analysis

- **Comparison with Newton-Raphson:** The FDLF algorithm's effectiveness will be evaluated by comparing its performance with the Newton-Raphson method in terms of computational speed, storage requirements, and accuracy. This includes comparing the expected voltages and angles with the findings received by power system simulation software (PSAF).
- **Scalability Analysis:** The project aims to analyze the algorithm's behavior as the number of buses in the power system varies, with a focus on its computational efficiency and stability. Power systems vary in size and complexity, requiring a load flow algorithm that is versatile enough to handle different scenarios efficiently.

3.2 Design Method (PO(a))

The design method involves applying engineering knowledge to develop the MATLAB code for the FDLF algorithm. Mathematical derivations and equations are provided to explain the implementation process.

The basis of FDLF approach has been explained such that:

1. The change in phase angle δ directly affects real power P , leaving the flow of reactive power Q relatively unchanged.
2. The change in voltage magnitude $|V|$ directly affects reactive power Q , leaving the flow of real power P relatively unchanged.

The two separate system of equations formed by incorporation of the approximations into the Jacobian submatrices for a 4-bus system^[1] are:

$$\begin{bmatrix} \frac{\partial P_2}{\partial \delta_2} & \cdots & \frac{\partial P_2}{\partial \delta_4} \\ \vdots & \mathbf{J}_{11} & \vdots \\ \frac{\partial P_4}{\partial \delta_2} & \cdots & \frac{\partial P_4}{\partial \delta_4} \end{bmatrix} \begin{bmatrix} \Delta \delta_2 \\ \vdots \\ \Delta \delta_4 \end{bmatrix} = \begin{bmatrix} \Delta P_2 \\ \vdots \\ \Delta P_4 \end{bmatrix}$$

and

$$\begin{bmatrix} |V_2| \frac{\partial Q_2}{\partial |V_2|} & \cdots & |V_4| \frac{\partial Q_2}{\partial |V_4|} \\ \vdots & \mathbf{J}_{22} & \vdots \\ |V_2| \frac{\partial Q_4}{\partial |V_2|} & \cdots & |V_4| \frac{\partial Q_4}{\partial |V_4|} \end{bmatrix} \begin{bmatrix} \frac{\Delta V_2}{|V_2|} \\ \vdots \\ \frac{\Delta V_4}{|V_4|} \end{bmatrix} = \begin{bmatrix} \Delta Q_2 \\ \vdots \\ \Delta Q_4 \end{bmatrix}$$

The system of equations obtained upon further calculations are:

$$\underbrace{\begin{bmatrix} -B_{22} & -B_{23} & -B_{24} \\ -B_{32} & -B_{33} & -B_{34} \\ -B_{42} & -B_{43} & -B_{44} \end{bmatrix}}_{\mathbf{B}} \begin{bmatrix} \Delta \delta_2 \\ \Delta \delta_3 \\ \Delta \delta_4 \end{bmatrix} = \begin{bmatrix} \frac{\Delta P_2}{|V_2|} \\ \frac{\Delta P_3}{|V_3|} \\ \frac{\Delta P_4}{|V_4|} \end{bmatrix}$$

and

$$\underbrace{\begin{bmatrix} -B_{22} & -B_{23} & -B_{24} \\ -B_{32} & -B_{33} & -B_{34} \\ -B_{42} & -B_{43} & -B_{44} \end{bmatrix}}_{\mathbf{B}} \begin{bmatrix} \Delta |V_2| \\ \Delta |V_3| \\ \Delta |V_4| \end{bmatrix} = \begin{bmatrix} \frac{\Delta Q_2}{|V_2|} \\ \frac{\Delta Q_3}{|V_3|} \\ \frac{\Delta Q_4}{|V_4|} \end{bmatrix}$$

The required system of equations to be solved using MATLAB are:

$$\begin{aligned} [\Delta P / |V|] &= [B'] [\Delta \delta] \\ [\Delta Q / |V|] &= [B''] [\Delta |V|] \end{aligned}$$

[1] Reference: John J. Grainger, William D. Stevenson Jr., "Power System Analysis", International Editions 1994.

The steps to perform iterative calculations can mentioned as follows:

1. Calculate the initial mismatches $\Delta P/|V|$.
2. Solve equation for $\Delta\delta$.
3. Update the angles δ and use them to calculate mismatches $\Delta Q/|V|$.
4. Solve equation for ΔV and update $|V|$.
5. Repeat the steps for next iteration until convergence.

3.3 Flowchart

The flowchart demonstrating Fast Decoupled Load Flow (FDLF) study algorithm has been shown below:

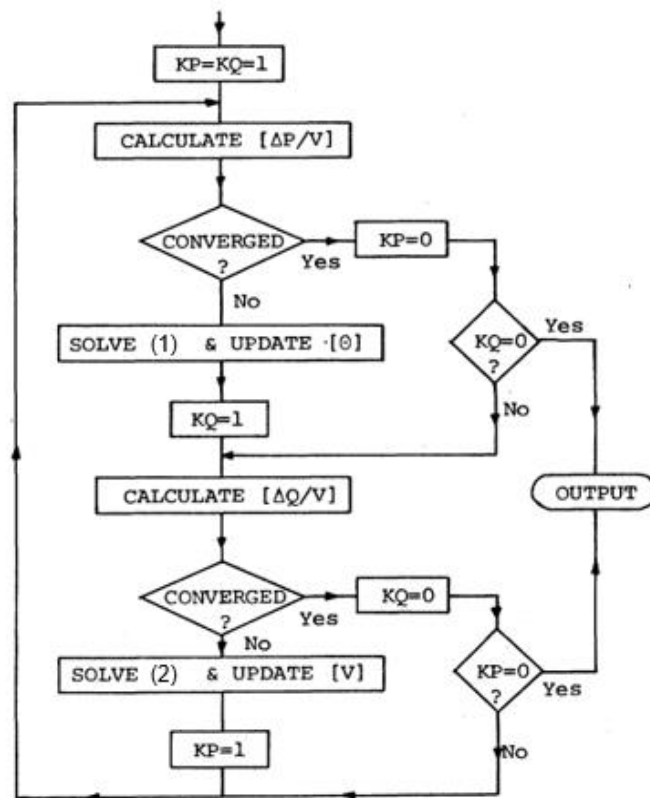


Figure: Flowchart of implemented algorithm

3.4 Full Source Code of Firmware

<pre> clc; clear all; close all; %% Reading from Excel fileName = 'IEEE_14.xlsx'; sheetName1 = 'LineData'; sheetName2 = 'BusData'; %Read line data from Excel lineData = xlsread(fileName,sheetName1); %Read bus data from Excel busData = xlsread(fileName,sheetName2); %% Work on line data nline = lineData(:,1); %Sl of line Sb = lineData(:,2); %Sending End Bus Rb = lineData(:,3); %Receiving End Bus R = lineData(:,4); %Resistance of line in pu X = lineData(:,5); %Reactance of line in pu B = 1i*lineData(:,6); %Capacitive Susceptance in pu t = lineData(:,7); %Nominal Tap Ratio Z = R + 1i*X; Y = 1./Z; bus = max(max(Sb),max(Rb)); %Number of buses line = length(nline); %Number of lines %% Fix the Ybus Ybus = zeros(bus,bus); %Initialise the Bus Admittance Matrix for k=1:line %Loop for Tap Ratio if t(k)~=1 t1 = (1-1/t(k)); %NTS t2 = -t1/t(k); %TS B(k) = t2*B(k); end end for k=1:line %Loop for Ybus p = Sb(k); q = Rb(k); Ybus(p,p) = Ybus(p,p) + Y(k)/t(k) + B(k); Ybus(q,q) = Ybus(q,q) + Y(k)/t(k) + B(k); Ybus(p,q) = Ybus(p,q) - Y(k); Ybus(q,p) = Ybus(p,q); end %% Work on bus data busnumber = busData(:,1); %Bus number type = busData(:,2); %Bus type Vmag = busData(:,3); %Voltage magnitude V = busData(:,3).*exp(1i*busData(:,4)); %Complex Form ANG = busData(:,4); %Bus angle PG = busData(:,5); %Generated real power QG = busData(:,6); %Generated reactive power PL = busData(:,7); %Load real power QL = busData(:,8); %Load reactive power BaseMVA = 100; Load = PL + 1i*QL; Qmin = busData(:,9)/BaseMVA; % Minimum Reactive Power Limit Qmax = busData(:,10)/BaseMVA; % Maximum Reactive Power Limit G = real(Ybus); % Conductance B = imag(Ybus); % Susceptance Psp = (PG-PL)/BaseMVA; %Scheduled real power Qsp = (QG-QL)/BaseMVA; %Scheduled reactive power Ssp = Psp + 1i*Qsp; %Scheduled complex power PV = find(type==2); %Generator Bus PQ = find(type==3); %Load Bus NS = [PV;PQ]; M = length(PV); %No. of PV bus N = length(PQ); %No. of PQ bus O = length(NS); dP = zeros(O,1); dQ = zeros(N,1); MV = [dP; dQ]; %Mismatch </pre>	<pre> PVnew= find(type==2); PQnew= find(type==3); dell = ANG*180/pi; for m=1:bus for n=1:bus I(m,n) = -(V(m)-V(n))*Ybus(m,n); end I(m,m) = Ybus(m,:)*V; end for m=1:bus for n=1:bus Pow_Flow(m,n) = V(m)*I(m,n)'; end Pow_Flow(m,m) = V(m)*I(m,m)'; end s1 = find(type==1); Slack_bus_power = Pow_Flow(s1,s1)*BaseMVA; for m=1:bus for n=1:bus Loss(m,n) = Pow_Flow(m,n) + Pow_Flow(n,m); end Loss(m,m)=0; end ste=clock; % Stop the iteration time clock lf = [type Vmag dell real(S).*100 Qc.*100]; %Load Flow Data Losses = real(Loss); Total_loop_loss=sum((sum(real(Loss))))/2*BaseMVA; %% Show the outputs disp('-----') disp(' FDLF Load-Flow Study'); disp('-----') disp(date); fprintf('Number of iterations : %d \n', iter); fprintf('Error tolerance considered : %d \n', tol); fprintf('Solution time : %g sec.\n\n', etime(ste,st)); fprintf(' Ybus Matrix:\n'); disp('====='); disp(Ybus); disp(" B'"); disp('====='); disp(B1) disp(' B''); disp('====='); disp(B11); fprintf(' BUS Data:\n'); disp('====='); disp(' bus type v delta Pg Qg Pl Ql Qmin Qmax'); disp('-----'); disp(busData); fprintf('Load Flow:\n'); disp('====='); disp(' Bus Type Vpu Angle P(MW) Q(MVAR)'); disp('-----'); disp(lf); fprintf('Power loss:%d MW\n \n',Total_loop_loss); %% Write in Excel columnHeadings = {'Bus Type','V(pu)','Angle','P(MW)','Q(MVAR)'}; decFormat = {'%.4f','%.4f','%.4f','%.4f','%.4f'}; dataTable = array2table(lf,'VariableNames',columnHeadings); outFileName = 'IEEE_14_Out.xlsx'; outSheet1 = 'LoadFlow'; %Format data upto 4 decimal places </pre>
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<pre> %% Load Flow iter=1; %Number of iterations tol=1; %Tolerance st=clock; %Stop the iteration time clock while (tol>1e-8 & iter<100) S = V.*conj(Ybus*V); %Since I=Ybus*V S1 = S - Ssp; Qc = imag(S); dP=real(S1(NS)); dQ=imag(S1(PQ)); B1= -imag(Ybus(NS,NS)); B11= -imag(Ybus(PQ,PQ)); invB1 = inv(B1); invB11 = inv(B11); dth=-invB1*dP; %Correction vector : theta dV=-invB11*dQ; %Correction vector : Voltage MV=[dP; dQ]; ANG(NS) = ANG(NS)+ dth; Vmag(PQ) = Vmag(PQ)+ dV; V=Vmag.*exp(1i*ANG); iter = iter+1; tol =max(abs(MV)); end </pre>	<pre> formattedData = arrayfun(@(x) sprintf('%0.4f', x), 1f, 'UniformOutput', false); %Write column headings xlswrite(outFileName, columnHeadings, outSheet1, 'A1'); %Write output data on table xlswrite(outFileName, formattedData, outSheet1,'A2'); </pre>
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Table: Source Code for the main program

4 Implementation

4.1 Description

The implementation portion of the project involves the development of a MATLAB code to execute the Fast-Decoupled Load Flow (FDLF) algorithm for power flow analysis in electrical systems. The code is designed to handle systems with any number of buses, making it useful for many applications in power system research.

The implementation process consists of the following essential steps:

- i) **Reading Input Data:** The MATLAB function reads input data from Excel files containing information about the system's line data and bus data. The line data includes information such as sending end bus, receiving end bus, resistance, reactance, and capacitance of the lines. The bus data includes characteristics such as bus number, bus type, voltage magnitude, bus angle, generated real and reactive power, and load real and reactive power.
- ii) **Processing Line Data:** The code processes the line data to calculate the nodal admittance matrix (Y_{bus}) of the system. This requires translating the line parameters into per-unit numbers and accounting for tap ratios if present.
- iii) **Processing Bus Data:** The code processes the bus data to initialize the voltage magnitudes and angles at each bus. It specifies the kind of each bus (e.g., slack bus, generator bus, load bus) to determine the required computations during the power flow analysis.

- iv) **Load Flow Calculation:** The crux of the system involves calculating the power flow equations iteratively using the FDLF algorithm. The program determines the real and reactive power mismatches at each bus, computes the correction vectors for voltage magnitudes and angles, and changes the bus voltages accordingly. The repeated process continues until the mismatches converge within a predetermined tolerance.
- v) **Analysis of Results:** Once the power flow calculations converge, the code examines the collected results. It compares the predicted voltages and angles with the expected values and rates the accuracy of the solution. Additionally, the algorithm computes the power flows, losses, and other pertinent metrics to assess the system's operational conditions.
- vi) **Output Generation:** Finally, the MATLAB function creates output data summarizing the power flow study results. This comprises the bus voltages, angles, real and reactive power injections, power flows, losses, and other pertinent information. The resulting data can be exported to an Excel file for additional analysis and reporting.

Overall, the implementation of the FDLF algorithm in MATLAB provides a strong tool for monitoring and studying power flow in electrical systems with flexibility and efficiency.

5 Design Analysis and Evaluation

5.1 Novelty

The novelty of this project lies in the implementation of the Fast-Decoupled Load Flow (FDLF) algorithm for power flow analysis in electrical systems through source coding. While power flow analysis is a well-established field, the use of the FDLF algorithm offers several advantages, including computational efficiency and versatility in handling systems with any number of buses. By developing a MATLAB code to execute the FDLF algorithm, the project introduces a practical and efficient tool for power flow analysis, particularly in large-scale systems where computational efficiency is crucial.

5.2 Design Considerations (PO(c))

Design considerations are very important to ensure the effectiveness and feasibility of a project. The design considerations for this project include evaluating the effectiveness of the used algorithm in predicting voltages and angles. The results obtained were compared with simulation outputs from PSAF to make sure that the designed solution is considerable. Moreover, the convergence and run time of the code were taken into account. As a results, the number of maximum possible iterations had been limited.

5.2.1 Considerations to public health and safety

The design considerations for this project prioritize public health and safety by assuring the accuracy and dependability of the power flow analysis results. By precisely forecasting voltages and angles in electrical systems, the FDLF algorithm aids in the safe and efficient functioning of power networks, lowering the danger of equipment failures and power outages.

5.2.2 Considerations to environment (PO(g))

While the primary focus of the project is on technical factors, the application of efficient power flow analysis techniques such as FDLF indirectly adds to environmental sustainability. By improving the functioning of power systems, the algorithm helps decrease energy losses and enhance overall system efficiency, therefore lowering the environmental impact associated with electricity generation and distribution.

5.3 Investigations (PO(d))

The investigation phase of the project involves a comprehensive study of the obtained output from the FDLF algorithm and its comparison with simulation outputs from established power system simulation software (PSAF). This includes the following steps:

5.3.1 Design of Experiment

The FDLF algorithm is testified against 4 standard buses – IEEE-4 Bus System, IEEE-13 Bus System, IEEE-14 Bus System and IEEE-30 Bus System. If the output is congruent to the values obtained from power system simulation software (PSAF) we can clarify the validity of this algorithm.

5.3.2 Data Collection

Bus Data and Line Data of 4 standard buses are collected to verify the algorithm –

1. IEEE-4 Bus
2. IEEE-13 Bus
3. IEEE-14 Bus
4. IEEE-30 Bus

IEEE – 4 Bus:

This bus has 1 slack bus, 1 generator bus and 2 load buses.

Line Diagram:

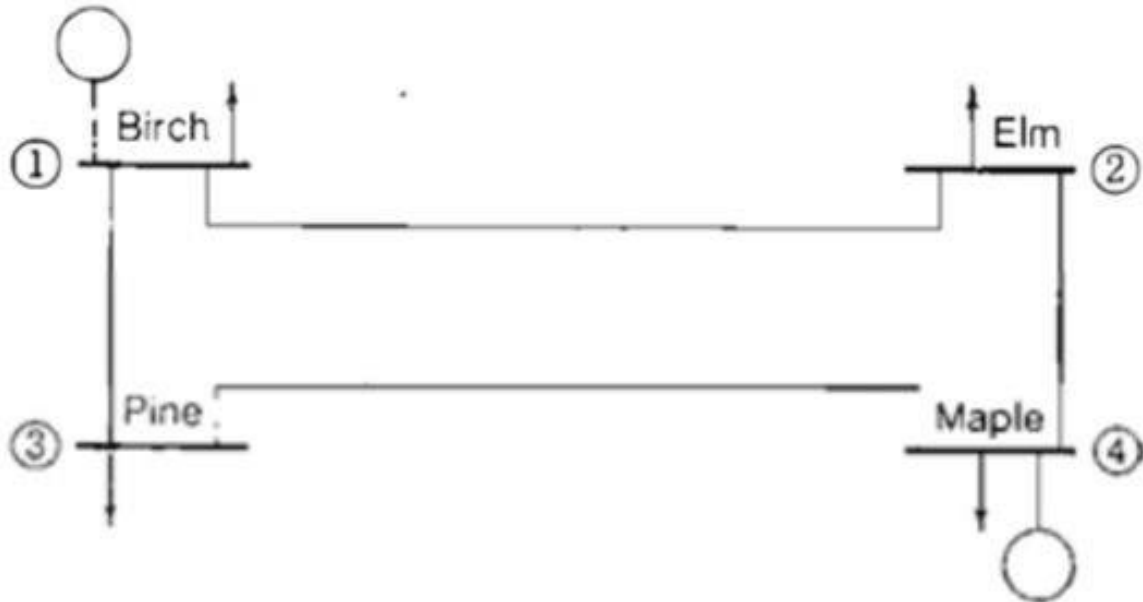


Fig: IEEE-4 Bus System Line Diagram

Line Data:

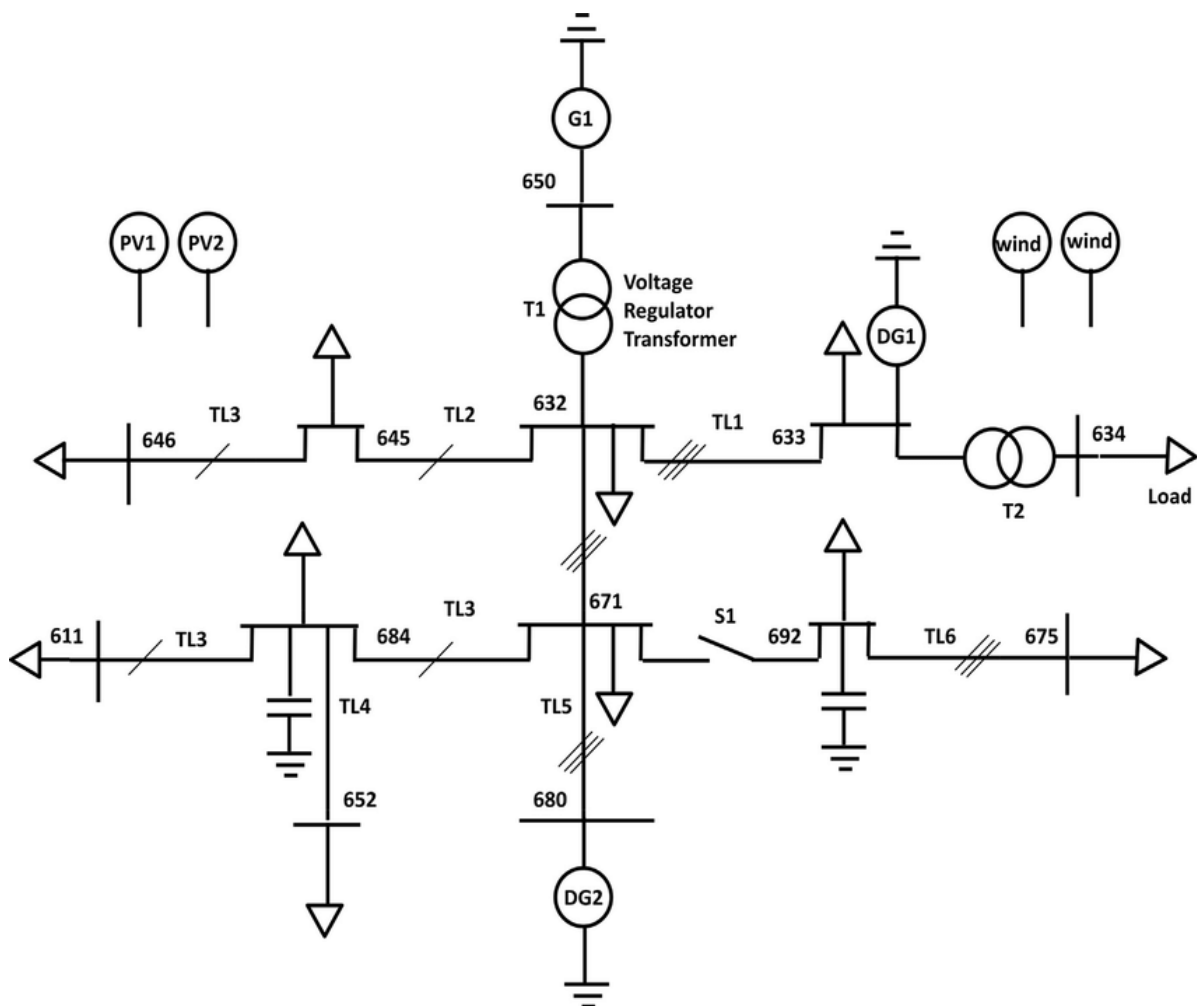
	A	B	C	D	E	F	G
1	Sl	Sb	Rb	R	X	B/2	Tap
2	1	1	2	0.01088	0.0504	0.05125	1
3	2	1	3	0.00744	0.0372	0.03875	1
4	3	2	4	0.00744	0.0372	0.03875	1
5	4	3	4	0.01272	0.0636	0.06375	1
6							

Bus Data:

	A	B	C	D	E	F	G	H	I	J
1	Bus	Type	V	Ang	Pg	Qg	Pl	Ql	Qmin	Qmax
2	1	1	1	0	0	0	50	30.99	0	0
3	2	3	1	0	0	0	170	105.35	0	0
4	3	3	1	0	0	0	200	123.94	0	0
5	4	2	1.02	0	318	0	80	49.58	-80	80

	ID	Zone	kV Base	V sol [pu]	Ang sol [deg]	Vmin [pu]	Vmax [pu]	P Gen [MW]	Q Gen [MVAR]	P Load [MW]	Q Load [MVAR]
1	B1	0	13.80	1.000	0.0	0.900	1.100	186.83	114.14	50.00	30.99
2	B2	0	13.80	0.982	-1.0	0.900	1.100	0.00	0.00	170.00	105.35
3	B3	0	13.80	0.969	-1.9	0.900	1.100	0.00	0.00	200.00	123.94
4	B4	0	13.80	1.020	1.5	0.900	1.100	318.00	181.80	80.00	49.58

Line Diagram:



Fast-Decoupled Load Flow (FDLF) Study for a System with Any Number of Bus

Line Data:

	A	B	C	D	E	F	G
1	Sl	Sb	Rb	R	X	B/2	Tap
2	1	1	2	0.01938	0.05917	0.0264	1
3	2	1	5	0.05403	0.22304	0.0246	1
4	3	2	3	0.04699	0.19797	0.0219	1
5	4	2	4	0.05811	0.17632	0.017	1
6	5	2	5	0.05695	0.17388	0.0173	1
7	6	3	4	0.06701	0.17103	0.0064	1
8	7	4	5	0.01335	0.04211	0	1
9	8	4	7	0.0075	0.0075	0	1
10	9	4	8	0.000375	0.012	0	1
11	10	5	6	0.002	0.024	0	1
12	11	6	10	0.09498	0.1989	0	1
13	12	6	11	0.12291	0.25581	0	1
14	13	6	12	0.06615	0.13027	0	1
15	14	7	8	0	0.11001	0	1
16	15	8	9	0.03181	0.0845	0	1
17	16	8	13	0.12711	0.27038	0	1
18	17	9	10	0.08205	0.19207	0	1
19	18	11	12	0.22092	0.19988	0	1
20	19	12	13	0.17093	0.34802	0	1

Bus Data:

	A	B	C	D	E	F	G	H	I	J
1	Bus	Type	V	Ang	Pg	Qg	Pl	Ql	Qmin	Qmax
2	1	1	1.06	0	0	0	0	0	-43.59	43.59
3	2	2	1.02	0	100	0	0	0	-40	50
4	3	3	1	0	0	0	95	20	0	0
5	4	3	1	0	0	0	45	0	0	0
6	5	3	1	0	0	0	10	2	0	0
7	6	3	1	0	0	0	11	7.5	0	0
8	7	3	1	0	0	0	12	5.8119	0	0
9	8	2	1	0	0	0	25	7	0	0
10	9	3	1	0	0	0	20	6	0	0
11	10	3	1	0	0	0	20	5	0	0
12	11	3	1	0	0	0	30	10	0	0
13	12	2	1.01	0	20	0	13.5	6	-60	100
14	13	3	1	0	0	0	20	5	0	0

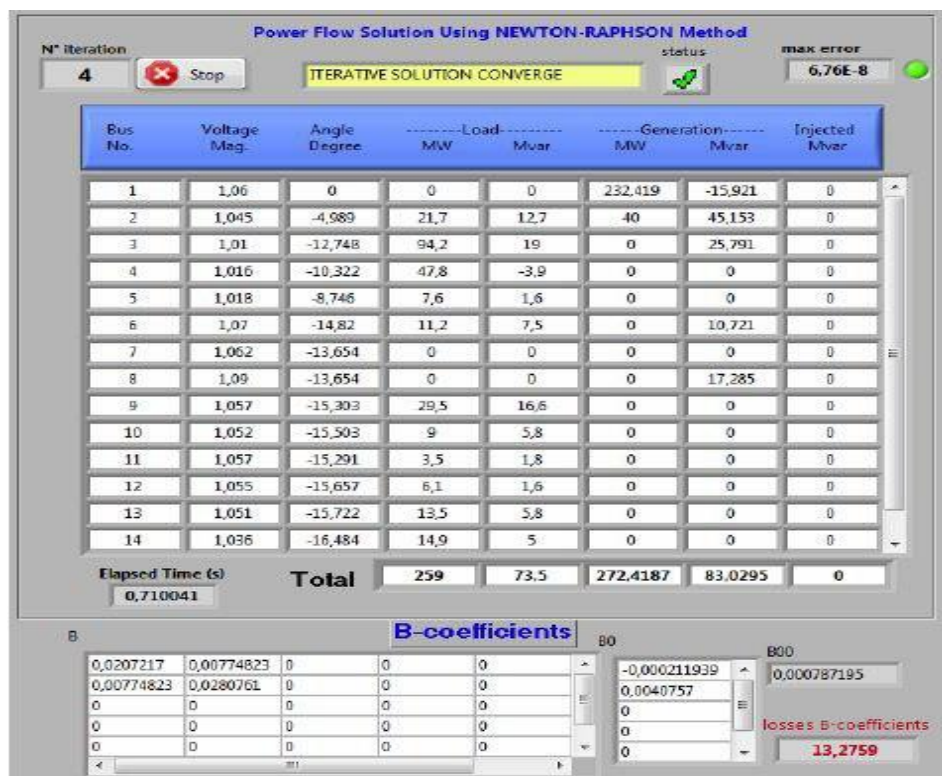
Simulation Output of PSAT:

	ID	Zone	kV Base	V sol [pu]	Ang sol [deg]	Vmin [pu]	Vmax [pu]	P Gen [MW]	Q Gen [MVAR]	P Load [MW]	Q Load [MVAR]	P motor [MW]	Q motor [MVAR]	P shunt [MW]	Q shunt [MVAR]
1	B1	0	138.00	1.060	0.0	0.900	1.100	199.72	53.04	0.00	0.00	0.00	0.00	0.00	0.00
2	B10	0	138.00	0.954	-12.2	0.900	1.100	0.00	0.00	20.00	5.00	0.00	0.00	0.00	0.00
3	B11	0	138.00	0.960	-12.8	0.900	1.100	0.00	0.00	30.00	10.00	0.00	0.00	0.00	0.00
4	B12	0	138.00	1.010	-12.5	0.900	1.100	20.00	54.04	13.50	6.00	0.00	0.00	0.00	0.00
5	B13	0	138.00	0.961	-13.3	0.900	1.100	0.00	0.00	20.00	5.00	0.00	0.00	0.00	0.00
6	B2	0	138.00	1.020	-3.4	0.900	1.100	100.00	4.80	0.00	0.00	0.00	0.00	0.00	0.00
7	B3	0	138.00	0.944	-12.2	0.900	1.100	0.00	0.00	95.00	20.00	0.00	0.00	0.00	0.00
8	B4	0	138.00	0.967	-10.7	0.900	1.100	0.00	0.00	45.00	0.00	0.00	0.00	0.00	0.00
9	B5	0	138.00	0.983	-9.3	0.900	1.100	0.00	0.00	10.00	2.00	0.00	0.00	0.00	0.00
10	B6	0	138.00	0.985	-10.2	0.900	1.100	0.00	0.00	11.00	7.50	0.00	0.00	0.00	0.00
11	B7	0	138.00	0.965	-10.7	0.900	1.100	0.00	0.00	12.00	5.81	0.00	0.00	0.00	0.00
12	B8	0	138.00	0.965	-11.0	0.900	1.100	0.00	0.00	25.00	7.00	0.00	0.00	0.00	0.00
13	B9	0	138.00	0.953	-12.0	0.900	1.100	0.00	0.00	20.00	6.00	0.00	0.00	0.00	0.00

Bus Data:

	A	B	C	D	E	F	G	H	I	J
1	Bus	Type	V	Ang	Pg	Qg	Pl	Ql	Qmin	Qmax
2	1	1	1.06	0	0	0	0	0	0	0
3	2	2	1.045	0	40	42.4	21.7	12.7	-40	50
4	3	2	1.01	0	0	23.4	94.2	19	0	40
5	4	3	1	0	0	0	47.8	-3.9	0	0
6	5	3	1	0	0	0	7.6	1.6	0	0
7	6	2	1.07	0	0	12.2	11.2	7.5	-6	24
8	7	3	1	0	0	0	0	0	0	0
9	8	2	1.09	0	0	17.4	0	0	-6	24
10	9	3	1	0	0	0	29.5	16.6	0	0
11	10	3	1	0	0	0	9	5.8	0	0
12	11	3	1	0	0	0	3.5	1.8	0	0
13	12	3	1	0	0	0	6.1	1.6	0	0
14	13	3	1	0	0	0	13.5	5.8	0	0
15	14	3	1	0	0	0	14.9	5	0	0

Simulation Output of LabView Simulation^[2]:



[2] Simulation results taken from https://www.researchgate.net/figure/Bus-data-of-IEEE-14-bus-system_tbl1_268924823

IEEE – 30 Bus:

Single Line Diagram:

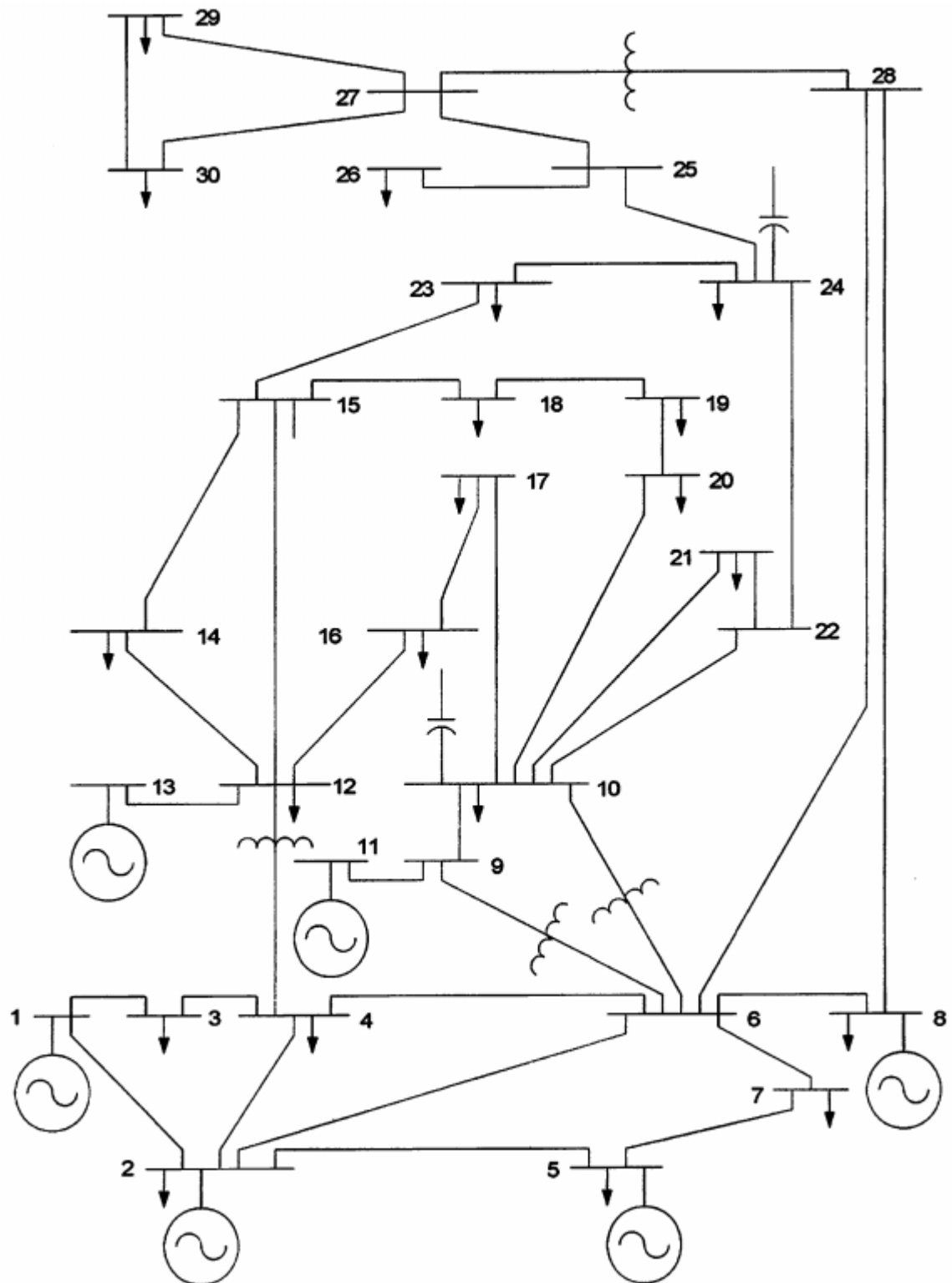


Fig: Single line diagram of 30-bus system

	A	B	C	D	E	F	G												
1	Sl	Sb	Rb	R	X	B/2	Tap		22	21	16	17	0.0824	0.1932	0	1			
2		1	1	2	0.0192	0.0575	0.0264	1	23	22	15	18	0.107	0.2185	0	1			23
3		2	1	3	0.0452	0.1852	0.0204	1	24	23	18	19	0.0639	0.1292	0	1			24
4		3	2	4	0.057	0.1737	0.0184	1	25	24	19	20	0.034	0.068	0	1			25
5		4	3	4	0.0132	0.0379	0.0042	1	26	25	10	20	0.0936	0.209	0	1			26
6		5	2	5	0.0472	0.1983	0.0209	1	27	26	10	17	0.0324	0.0845	0	1			27
7		6	2	6	0.0581	0.1763	0.0187	1	28	27	10	21	0.0348	0.0749	0	1			28
8		7	4	6	0.0119	0.0414	0.0045	1	29	28	10	22	0.0727	0.1499	0	1			29
9		8	5	7	0.046	0.116	0.0102	1	30	29	21	23	0.0116	0.0236	0	1			30
10		9	6	7	0.0267	0.082	0.0085	1	31	30	15	23	0.1	0.202	0	1			31
11		10	6	8	0.012	0.042	0.0045	1	32	31	22	24	0.115	0.179	0	1			32
12		11	6	9	0	0.208	0	1.0155	33	32	23	24	0.132	0.27	0	1			33
13		12	6	10	0	0.556	0	0.9629	34	33	24	25	0.1885	0.3292	0	1			34
14		13	9	11	0	0.208	0	1	35	34	25	26	0.2544	0.38	0	1			35
15		14	9	10	0	0.11	0	1	36	35	25	27	0.1093	0.2087	0	1			36
16		15	4	12	0	0.256	0	1.0129	37	36	28	27	0	0.369	0	0.9581			37
17		16	12	13	0	0.14	0	1	38	37	27	29	0.2198	0.4153	0	1			38
18		17	12	14	0.1231	0.2559	0	1	39	38	27	30	0.3202	0.6027	0	1			39
19		18	12	15	0.0662	0.1304	0	1	40	39	29	30	0.2399	0.4533	0	1			40
20		19	12	16	0.0945	0.1987	0	1	41	40	8	28	0.0636	0.2	0.0214	1			41
21		20	14	15	0.221	0.1997	0	1	42	41	6	28	0.0169	0.0599	0.065	1			42

A	B	C	D	E	F	G	H	I	J	K
1	1	1.06	0	0	0	0	0	0	0	
2	2	1.043	0	40	48.8	21.7	12.7	-40	50	
3	3	1	0	0	0	2.4	1.2	0	0	
4	3	1.06	0	0	0	7.6	1.6	0	0	
5	2	1.01	0	0	36	94.2	19	-40	40	
6	3	1	0	0	0	0	0	0	0	
7	3	1	0	0	0	22.8	10.9	0	0	
8	2	1.01	0	0	30.8	30	30	-10	40	
9	3	1	0	0	0	0	0	0	0	
10	3	1	0	0	19	5.8	2	0	0	
11	2	1.082	0	0	16.1	0	0	-6	24	
12	3	1	0	0	0	11.2	7.5	0	0	
13	2	1.071	0	0	10.4	0	0	-6	24	
14	3	1	0	0	0	6.2	1.6	0	0	
15	3	1	0	0	0	8.2	2.5	0	0	
16	3	1	0	0	0	3.5	1.8	0	0	
17	3	1	0	0	0	9	5.8	0	0	
18	3	1	0	0	0	3.2	0.9	0	0	
19	3	1	0	0	0	9.5	3.4	0	0	
20	3	1	0	0	0	2.2	0.7	0	0	
21	3	1	0	0	0	17.5	11.2	0	0	
22	3	1	0	0	0	0	0	0	0	
23	3	1	0	0	0	3.2	1.6	0	0	
24	3	1	0	0	4.3	8.7	6.7	0	0	
25	3	1	0	0	0	0	0	0	0	
26	3	1	0	0	0	3.5	2.3	0	0	
27	3	1	0	0	0	0	0	0	0	
28	3	1	0	0	0	0	0	0	0	
29	3	1	0	0	0	0	0	0	0	
30	3	1	0	0	0	21.7	12.7	0	0	

Simulation Output^[3]:

Table 1: Load bus data for IEEE 30 bus system using NR method

Bus	Voltage		Generation		Load		Injected
	Mag	Angle (degree)	P (MW)	Q (Mvar)	P (MW)	Q (Mvar)	Q (Mvar)
1	1.06	0	2.61	-0.17	0	0	0
2	1.043	-5.497	0.4	0.488	-0.217	-0.127	0
3	1.022	-8.004	0	0	-0.024	-0.012	0
4	1.013	-9.661	0	0	-0.076	-0.016	0
5	1.01	-14.381	0	0.36	-0.942	-0.19	0
6	1.012	-11.398	0	0	0	0	0
7	1.003	-13.15	0	0	-0.228	-0.109	0
8	1.01	-12.115	0	0.308	-0.3	-0.3	0
9	1.051	-14.434	0	0	0	0	0
10	1.044	-16.024	0	0	-0.058	-0.02	0.19
11	1.082	-14.434	0	0.161	0	0	0
12	1.057	-15.302	0	0	-0.112	-0.075	0
13	1.071	-15.302	0	0.104	0	0	0
14	1.042	-16.191	0	0	-0.062	-0.016	0
15	1.038	-16.278	0	0	-0.082	-0.025	0
16	1.045	-15.88	0	0	-0.035	-0.018	0
17	1.039	-16.188	0	0	-0.09	-0.058	0
18	1.028	-16.884	0	0	-0.032	-0.009	0
19	1.025	-17.052	0	0	-0.095	-0.034	0
20	1.029	-16.852	0	0	-0.022	-0.007	0
21	1.032	-16.468	0	0	-0.175	-0.112	0
22	1.033	-16.455	0	0	0	0	0
23	1.027	-16.662	0	0	-0.032	-0.016	0
24	1.022	-16.83	0	0	-0.087	-0.067	0.043
25	1.019	-16.424	0	0	0	0	0
26	1.001	-16.842	0	0	-0.035	-0.023	0
27	1.026	-15.912	0	0	0	0	0
28	1.011	-12.057	0	0	0	0	0
29	1.006	-17.136	0	0	0	0	0
30	0.995	-18.015	0	0	-0.217	-0.127	0

[3] Source: <https://www.wseas.org/multimedia/journals/power/2018/a125916-585.pdf>

Output of FDLF Algorithm:

IEEE – 4 Bus:

Simulation Output of MATLAB:

	A	B	C	D	E
1	Bus Type	V(pu)	Angle	P(MW)	Q(MVAR)
2	1	1	0	136.8294	83.1545
3	3	0.9823	-0.9686	-170	-105.35
4	3	0.969	-1.8705	-200	-123.94
5	2	1.02	1.5277	238	132.2204

IEEE -13 Bus:

Simulation Output of MATLAB:

	A	B	C	D	E
1	Bus Type	V(pu)	Angle	P(MW)	Q(MVAR)
2	1	1.06	0	198.6007	45.3492
3	2	1.02	-3.3215	100	-29.8649
4	3	0.9605	-12.2084	-95	-20
5	3	0.9958	-10.9339	-44.9999	0
6	3	1.0007	-9.3509	-10	-2
7	3	0.9996	-10.212	-11	-7.5
8	3	0.9945	-11.0038	-12.0001	-5.8119
9	2	1	-11.3154	-25	52.9
10	3	0.985	-12.2112	-20	-6
11	3	0.9774	-12.2479	-20	-5
12	3	0.9675	-12.571	-30	-10
13	2	1.01	-11.9155	6.5	27.2535
14	3	0.9814	-13.133	-20	-5

IEEE-14 Bus:**Simulation Output of MATLAB:**

	A	B	C	D	E
1	Bus Type	V(pu)	Angle	P(MW)	Q(MVAR)
2	1	1.06	0	232.9702	-13.9429
3	2	1.045	-5.0017	18.3	39.8668
4	2	1.01	-12.7891	-94.2	11.647
5	3	1.0084	-10.1865	-47.8	3.9
6	3	1.0136	-8.7456	-7.6	-1.6
7	2	1.07	-14.985	-11.2	89.4749
8	3	1.0261	-13.2942	0	0
9	2	1.09	-13.2942	0	39.5367
10	3	1.0089	-14.9276	-29.5	-16.6
11	3	1.012	-15.2192	-9	-5.8
12	3	1.037	-15.2082	-3.5	-1.8
13	3	1.0516	-15.8237	-6.1	-1.6
14	3	1.0435	-15.809	-13.5	-5.8
15	3	1.0054	-16.3694	-14.9	-5

IEEE – 30 Bus:**Simulation Output of MATLAB:**

1	Bus Type	V(pu)	Angle	P(MW)	Q(MVAR)	17	3	1.0171	-17.2412	-3.5	-1.8
2	1	1.06	0	273.4422	-21.564	18	3	1.0084	-17.5709	-9	-5.8
3	2	1.043	-5.7526	18.3	35.9142	19	3	0.9983	-18.2909	-3.2	-0.9
4	3	1.0256	-8.4818	-2.4	-1.2	20	3	0.9947	-18.4761	-9.5	-3.4
5	3	1.018	-10.2404	-7.6	-1.6	21	3	0.9983	-18.2673	-2.2	-0.7
6	2	1.01	-14.8142	-94.2	17.8241	22	3	0.9952	-17.9835	-17.5	-11.2
7	3	1.0104	-11.981	0	0	23	3	0.9955	-17.8706	0	0
8	3	1.0025	-13.672	-22.8	-10.9	24	3	0.9944	-18.0019	-3.2	-1.6
9	2	1.01	-12.7546	-30	12.6125	25	3	0.9727	-18.3395	-8.7	-2.4
10	3	1.0331	-15.5285	0	0	26	3	0.9303	-18.178	0	0
11	3	1.0125	-17.4	-5.8	17	27	3	0.9109	-18.6817	-3.5	-2.3
12	2	1.082	-15.5285	0	25.4382	28	3	0.9134	-17.8087	0	0
13	3	1.0333	-16.6388	-11.2	-7.5	29	3	0.9952	-12.5508	0	0
14	2	1.071	-16.6388	0	28.8273	30	3	0.8605	-19.6648	0	0
15	3	1.0164	-17.5776	-6.2	-1.6	31	3	0.8038	-21.9671	-21.7	-12.7
16	3	1.01	-17.6389	-8.2	-2.5						

5.3.3 Results and Analysis

Accuracy of Results:

IEEE-4 Bus:

- Deviation of voltage was less than 0.001pu and deviation of angle was maximum 0.03 degree.
- There was no deviation in P and Q upto 4th digit.

IEEE-13 Bus:

- Deviation of voltage was less than 0.005pu and deviation of angle was maximum 0.2 degree.
- Deviation of P was 0.12MVA and deviation of Q was 4 MVAR.

IEEE-14 Bus:

- Deviation of voltage was less than 0.002pu and deviation of angle was maximum 0.03 degree.
- Deviation of P was 0.5MVA and deviation of Q was 1 MVAR.

IEEE-30 Bus:

- Deviation of voltage was less than 0.01pu and deviation of angle was maximum 1 degree.
- Deviation of P was negligible but the deviation of Q was high.

Computational Efficiency:

For 4-bus system, run time of the code was 0.001s which was same for Newton Raphson method. For 13-bus system, run time of the code was 0.02 s which was 0.5s for Newton Raphson method. For 14-bus system, run time was 0.024s and for 30-bus system, run time was 0.12s. For Newton Raphson method time was 0.6s and 1.5s respectively.

Result shows that, computational efficiency has increased significantly for higher number of buses.

5.3.4 Interpretation and Conclusions on Data

- FDLF algorithm can effectively predict voltages and angles.
- With increase in bus number, the accuracy of algorithm reduces for reactive power calculations. Real power calculations do not deviate as much.
- The algorithm does not diverge in any case of tested standard buses.
- The deviations are mainly due to the approximations made in Fast Decoupled Load Flow study, compared to Newton-Raphson method.

5.4 Limitations of Tools (PO(e))

While the FDLF technique implemented in MATLAB offers computational efficiency and versatility, it has significant restrictions that need to be considered:

1. The method relies on certain simplifying assumptions and approximations, which may impair its accuracy in certain cases.
2. The computing speed of the technique may be impaired when studying extremely large-scale systems with a high number of buses and complex network configurations.

3. The correctness of the results derived from the FDLF algorithm may depend on the quality and completeness of the input data, including line and bus parameters.

Despite these limitations, the FDLF algorithm remains a powerful tool for power flow analysis, offering a balance between computational efficiency and accuracy for a wide range of practical applications.

5.5 Impact Assessment (PO(f))

Societal Impact: The FDLF method contributes to the efficient operation and planning of power systems. By ensuring reliable and efficient power flow, it indirectly benefits society by supporting the stable operation of electrical grids. This stability is crucial for the functioning of various societal infrastructures like hospitals, schools, businesses, and homes.

Cultural Impact: While the FDLF method doesn't directly impact culture, the broader field of electrical engineering, which includes techniques like FDLF, has shaped our modern lifestyle and culture by enabling technological advancements.

Health and Safety Impact: The FDLF method contributes to the safety of power systems by providing efficient and reliable load flow solutions. A stable power system is crucial for the operation of health facilities and emergency services. Moreover, by helping to prevent blackouts, it ensures the safety of electrical devices and reduces the risk of accidents caused by power failures.

Legal Impact: While the FDLF method itself doesn't have direct legal implications, the power industry is subject to various regulations and standards to ensure safety, reliability, and fairness. The use of scientifically sound methods like FDLF can help power companies comply with these regulations by ensuring accurate and efficient load flow analysis.

These impacts are indirect and pertain to the role of the FDLF method in power system analysis and operation. The method itself is a technical tool used by engineers and doesn't directly interact with societal or cultural aspects. It's also worth noting that the information available on the direct impact of the FDLF method on health and safety, and legal issues is limited.

5.6 Sustainability Evaluation

Sustainability Considerations for Fast-Decoupled Load Flow Study Project:

1. Energy Efficiency in Computational Resources:

Sustainable computing practices involve optimizing algorithms to minimize energy consumption during execution. The Fast-Decoupled Load Flow (FDLF) algorithm, with its emphasis on computational efficiency and reduced storage requirements, aligns with energy-conscious computing. This sustainability aspect is crucial in promoting environmentally friendly practices within the realm of power system analysis.

2. Reduction in Carbon Footprint:

As power systems continue to grow, the demand for load flow studies increases, impacting computational resources and, consequently, the carbon footprint of data centers. The FDLF algorithm, designed for efficiency, contributes to the reduction of carbon emissions associated with power-hungry computational processes. This aligns with broader sustainability goals in mitigating the environmental impact of technology.

3. Versatility and Long-Term Applicability:

Sustainability also involves creating solutions with long-term applicability. The FDLF algorithm's versatility in handling networks of any size ensures its usefulness in evolving power systems. By providing a sustainable solution adaptable to varying bus numbers, the project contributes to the longevity and relevance of load flow studies in power engineering.

4. Resource Conservation through GUI Design:

The inclusion of a Graphical User Interface (GUI) in the project promotes user-friendly interactions, potentially reducing the need for extensive training and documentation. This approach aligns with sustainability by conserving human resources, as efficient GUIs contribute to a more intuitive and accessible user experience.

5. Open-Source Collaboration for Continuous Improvement:

Open-source development encourages collaboration and knowledge sharing among researchers and practitioners globally. By making the FDLF algorithm and associated tools open source, the project fosters a community-driven approach to continuous improvement and refinement. This collaborative model contributes to the sustainable evolution of load flow analysis methods.

6. Integration with Renewable Energy Systems:

Sustainability in power systems goes beyond computational efficiency. The FDLF algorithm's application can extend to the analysis of renewable energy integration. By ensuring accurate and efficient load flow studies, the project indirectly supports the sustainable integration of renewable energy sources into the power grid.

In summary, the Fast-Decoupled Load Flow (FDLF) Study Project incorporates sustainability considerations through its focus on energy-efficient algorithms, reduction of carbon footprint, long-term applicability, resource conservation through GUI design, open-source collaboration, and support for renewable energy integration. These aspects align the project with broader sustainability goals, contributing to environmentally conscious and socially responsible advancements in power system analysis.

5.7 Ethical Issues

1. Credit for Intellectual Resources:

The project team acknowledges the influence of external sources, such as academic papers and websites, by crediting them appropriately. This practice demonstrates intellectual honesty and ensures transparency about the foundations of the project.

2. Implementing Own Code:

The team affirms the originality of their work by stating that the project's code was independently developed, not copied from the internet or previous projects. This declaration underscores the team's commitment to academic integrity and innovation.

3. Uniform Work Distribution:

Acknowledging the collaborative nature of the project, the team ensures fair task allocation among members. This approach prevents any individual from being overburdened or having insufficient responsibilities, fostering a balanced and inclusive team dynamic.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

The team consisted of five members, each contributing unique skills and expertise to the project:

Student Name	Contribution
Md. Muaz Rahman (2006089)	- Implementation of Excel file handling codes in MATLAB. - Debugging and optimization.
Mohammad Al Hosan (2006090)	- Design and implementation of bus and line data manipulation code in MATLAB. - Analysis of results.
Md. Nakib (2006091)	- Formulation of problem statement. - Data collection and coding of load flow algorithm.
Md. Samiul Hossen Ranjy (2006092)	- Literature review. - Analysis of existing algorithms. - Interpretation of results.
Dipto Archishman Saha (2006093)	- Integration of MATLAB code for FDLF algorithm implementation. - Data processing from Excel files.

6.2 Mode of Team Work

The team took a collaborative strategy to achieve good teamwork throughout the project. Regular meetings were held to discuss project progress, allocate work, and handle any issues discovered. Communication channels such as email, messaging apps, and video conversations were employed for smooth cooperation, especially during distant work periods.

The team created clear roles and duties for each member, ensuring that tasks were split equitably based on individual skills and expertise. Collaborative solutions such as Google Drive and GitHub were leveraged for sharing documents, code repositories, and project resources, promoting effective collaboration and version control.

Feedback and constructive criticism were encouraged within the team to foster ongoing progress and assure the quality of outputs. Regular progress reports were delivered during meetings to track project milestones and address any issues or complaints promptly.

6.3 Diversity Statement of Team

The team consisted of persons with varied backgrounds, experiences, and opinions, contributing to a rich and dynamic collaboration atmosphere. Each member provided distinct skills and knowledge to the project, improving the overall efficacy and originality of the team.

Diversity among the team stimulated invention and creativity, allowing for a full investigation of the problem statement and the development of viable solutions. Different cultural origins and opinions strengthened discussions and decision-making processes, resulting to well-rounded outcomes and solutions.

The team accepted diversity as a strength, harnessing the collective talents and insights of its members to confront difficulties and achieve project objectives successfully.

6.4 Log Book of Project Implementation

Date	Milestone achieved	Comments
20/12/2023	Project Proposal	Proposal was accepted
10/01/2024	Project Update -1 (Online)	Algorithm and flowchart were developed, update was shown through presentation slides
07/02/2024	Project Update -2 (Offline)	Implementation of full code, results were shown through presentation slides
25/02/2024	Final Demonstration	Demonstration of complete project including presentation slides, submission of report on project

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

FOR IMMEDIATE RELEASE

New Breakthrough: Fast Decoupled Load Flow Study for Any Bus System

Dhaka, February 07, 2024 – A significant moment in power systems study has been reached with the launch of a remarkable project. Presenting the "Fast Decoupled Load Flow Study for a System with Any Number of Buses." The current method, which offers improved computational efficiency and minimized storage requirements, simplifies power flow analysis. It has been developed by a group of students.

Large-scale systems present challenges for traditional approaches, but this effort overcomes them. Regardless of the complexity of the system, its flexible structure guarantees quick analysis, giving engineers useful information for improved grid stability and optimization.

This project aims to hallmark new dimensions in power engineering.

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8 Future Work (PO(l))

In future work, numerous options could be pursued to enhance and extend the scope of this project:

Integration of complex methods: Explore and construct more complex power flow analysis methods, such as the Newton-Raphson method or the Gauss-Seidel method, to compare their performance with the FDLF algorithm in different scenarios.

Optimization strategies: Investigate optimization strategies to boost the computational efficiency of the developed MATLAB code, enabling faster execution and analysis of larger-scale power systems.

Graphical Visualization: Develop graphical visualization tools to show the results of power flow analysis, including bus voltages, angles, and power flows, in an intuitive and user-friendly manner.

Integration with Real-Time Data: Integrate the MATLAB code with real-time data collecting devices to permit dynamic power flow analysis and monitoring of electrical grids in real-world scenarios.

Validation and Testing: Conduct extensive validation and testing of the implemented MATLAB code using different test cases and benchmarking against established power system simulation tools to ensure accuracy and reliability.

9 References

- [1] John J. Grainger, William D. Stevenson Jr., “Power System Analysis”, International Editions 1994.
- [2] B. Stott, O. Alsac, “Fast Decoupled Load Flow”, DOI:
<https://ieeexplore.ieee.org/document/4075431>
- [3] A. Monticelli, A. Garcia, O. R. Saavedra, “Fast Decoupled Load Flow: Hypothesis, Derivations, and Testing”, DOI:
https://www.academia.edu/27006861/Fast_decoupled_load_flow_hypothesis_derivations_and_testing
- [4] Yubin Yao, Mingwu Li, “Designs of Fast Decoupled Load Flow for Study Purpose”, DOI:
https://www.researchgate.net/publication/257711931_Designs_of_Fast_Decoupled_Load_Flow_for_Study_Purpose
- [5] Article: Decoupled Load Flow Methods (DLF), DOI:
<https://www.eeeguide.com/decoupled-load-flow-methods-dlf/>